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RESEARCH ARTICLE

Zero-Touch Network Security (ZTNS): A Network Intrusion Detection System Based on Deep Learning

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ABSTRACT The rapid evaluation of smart cities has revolutionized the research and development field to a very extensive level which presents challenges in handling massive amounts of data. However, the integration of IoT into various aspects of life has introduced various challenges related to the security and privacy of IoT systems. IoT sensors capture large volumes of sensitive customer data, which can potentially make them a target and pose serious threats, including financial loss and identity theft. Strong intrusion detection systems are essential for protecting networked, data-driven ecosystems from potential cyber threats. In this paper, we propose a novel deep learning-based approach that focuses on emerging zero-touch networks that autonomously manage network resources to ensure network security, the proposed approach identifies various network intrusions such as DDoS, Botnet, Brute force, and Infiltration. Our proposed approach presents a major improvement in IoT security. We have used the CICIDS-2018 benchmark dataset and propose a deep learning-based network intrusion detection System for Zero Touch Networks (DL-NIDS-ZTN). The proposed study utilizes convolutional neural networks that correctly identify benign and malicious traffic and achieve 99.80% accuracy with the CICIDS-2018 dataset. By implementing the DL-NIDS-ZTN methodology, we aim to strengthen the security framework of smart cities and ensure the secure and seamless integration of IoT.

INDEX TERMS Intrusion detection, zero-touch networks, smart city, IoT, deep learning, convolutional neural networks.

I. INTRODUCTION

The rise of smart cities in various sectors such as smart grids, smart water treatment facilities, and smart homes, presented new security challenges, especially cyber-attacks. By incorporating modern innovations like sensors and smart meters to improve various services, cities are becoming smarter. Unfortunately, because of their correlation, these devices also make targets for attackers. Every cyberattack impacts economic losses and legal significance in smart cities, hence network development of intrusion detection systems (IDS) has a global impact on smart cities. Network Intrusion Detection System (NIDS) is an effective security tool for the detection and prevention of cyber-attacks on any network or system. NIDS is responsible for the discovery of illegal activities and the protection of network infrastructure against cyberattacks, as well as the reduction of financial and operational losses [1]. The incredible growth in network traffic

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asks for the inquiry of security risks, however, it has made it difficult for NIDS systems to effectively identify intrusions. There are still several prospects to examine intrusions with the help of NIDS to effectively detect network attackers. The research on the application of deep learning approaches for NIDS is now in its early stages. These developments do, however, raise the possibility of cyberattacks that might bring these systems to a slowdown. Based on the literature, a network intrusion detection system is categorized into three groups based on the network architecture: (i) network-based ID systems [2], which analyze the components of distinctive packets to identify malicious anomaly system traffic signature patterns; (ii) server signature ID systems, which analyze the individual hosts' activity system logs to identify malicious attacks; and (iii) hybrid ID systems [3], which incorporate both anomaly and network signature-based intrusion detection. The [4] research, presented host-based features and an automatic network-based features integrated intrusion detection scheme to enhance the network attack detection performance. They used the SCVIC-CIDS-2021 dataset which is obtained from the network packets integration and host logs of the CSE-CIC-IDS2018 dataset. Additionally, they utilized the Gated Recurrent Unit (GRU) and Auto-encoder (AE) for feature derivation to minimize the dimensionality mismatch among host-based and network-based features. It improves the accuracy and efficiency of identifying and stopping malicious or unauthorized activities. Recently, the Zero-Touch Network(ZTN) has emerged to automatically coordinate and manage network resources. ZTN is necessary for operators to improve performance, and minimize the time for new features. It is an adaptive network that can adapt and modify based on the signals in the data that they gather, evaluate network activity, ensuring network security and privacy. Currently, there are numerous approaches available that have been applied to the detection of intrusions in smart cities-enabled zero-touch networks. These attempts utilize machine learning techniques, which are incorporated into deep learning models.

However, the full potential of applying deep learning techniques in zero-touch networks to improve intrusion detection has not yet been realized. The majority of proposed approaches have focused on triggering attack alerts in traffic data; however, it is vital to know the address of the attacker to avoid and block such attacks. Thus, one of the top problems for researchers today is keeping these devices secure. IoT is an area that is rapidly growing, thus it is important to conduct ongoing research and development to stay in touch with new standards, technologies, and best practices. IoT technological developments have sparked the development of cutting-edge applications in smart cities. A few examples of how IoT is changing conventional networking applications are smart grids, remote healthcare monitoring, autonomous vehicles, and industrial automation. IoT systems frequently consist of several connected devices and parts, which complicates the overall design. This complexity might result in a large number of potential entry points for attackers, each requiring security consideration. According to the literature, there is a need for deep learning-based intrusion detection systems that can work ideally in the identification of zero-touch networks. The researchers have investigated utilizing machine learning (ML) and deep learning (DL) approaches to meet the requirements of a successful IDS. These methods have gained a lot of power in the field of network security as a result of the ten-year rise in the capability of graphics processing units. In competition, IoT systems must constantly adapt and survive in a precise and predictable manner with safety as a key priority. Anomaly detection approaches create a baseline of typical network behavior rather than merely relying on known attack patterns. Changes from this norm can be a sign of forthcoming attacks, particularly zero-touch ones. In this proposed research study, our main objective is to automatically detect and stop lethal actions occurring within these networks, with a concentration on assaults that are directed against the network itself, also known as NIDS attacks. The model is capable of distinguishing between different types of network traffic, classifying them, and allowing only normal traffic to flow within the smart city. The proposed intrusion detection model using deep learning for zero-touch networks in smart cities, may recognize malicious traffic and find various NIDS attacks. Detecting harmful traffic inside a zero-touch network makes it possible to identify numerous variations in traffic, which improves performance and lets only legitimate traffic through the system with the help of DL-based NIDS. Smart cities, where smart meters and sensors produce large volumes of data, are the major causes of network attacks. Using the proposed deep learning-based model for NIDS adds an advantage to zero-touch networks for smart cities and makes the detection of intrusions more reliable and effective. Hence, this research has made the following contributions:

i. The proposed Deep Learning-based Network Intrusion Detection System with a secure, efficient, and resilient Zero-Touch Networks (DL-NIDS-ZTN) robust model is developed to classify whether the network traffic behavior is malicious or normal.
ii. The proposed model employs a multi-layered deep learning approach named CNNs which refers to a network that has 5 layers, including the dropout layers, max-pooling layer(for downsampling), flattening layer, fully connected layers, and an output layer for intrusion detection, the model correctly identifies the attacks in 99.80% of instances.
iii. The detailed analysis of the CICIDS-2018 dataset [40] dataset that is used for experimentation purposes in this study. This dataset is widely used for evaluating the effectiveness of Intrusion Detection Systems (IDS).
iv. The comparison of our proposed DL-NIDS-ZTN model with the existing approaches is based on achieved matrics.

The advent of smart cities has led to a rapid evolution in research and development, emphasizing the need to handle massive amounts of data efficiently. The integration of IoT into various aspects of life has, however, brought forth challenges related to the security and privacy of IoT systems. The proposed deep learning-based approach addresses these challenges by specifically targeting emerging zero-touch networks, which autonomously manage network resources to ensure network security. Zero-touch networks play a crucial role in smart cities, automatically coordinating and managing network resources to ensure quality of services. The performance of the proposed framework is validated by employing a benchmark CICIDS-2018 dataset. The experimental results demonstrated the framework's capacity to detect various types of cyber-attacks i.e., distributed Denial of Service (DDoS) attacks, Botnet, Brute Force, and Infiltration where the proposed model is capable of boosting the effectiveness of the classification model.

The remaining sections of this work are structured as follows: Section II explains the background and related work. Section III discusses the training models and dataset experiments for deep learning-based zero-touch networks for smart city environments. The methodology for NIDS is presented in Section IV, followed by a summary of the conclusions in Section V.

II. LITERATURE REVIEW

The impact of smart cities on the quality of life has been significant. The smart city is about managing everyday activities utilizing mobile applications and connecting everything citizens require for simple access and management [5]. The achievement of a smart city's objectives is dependent on several elements and challenges. Among the most important considerations and obstacles is network intrusion detection. To safeguard zero-touch network-enabled smart cities from cyberattacks, a comprehensive deep learning-based security framework is required. The primary objective of this study is to investigate how machine learning techniques can be applied to zero-touch networks. Due to technological advancements, a vast amount of data must be processed, and machine-learning techniques are used to discover useful patterns. These patterns could aid in recognizing the various forms of attacks that affect smart city systems. DDoS attacks are prevalent cyber-attack strategies that target service availability in smart cities, including flooding systems with many requests to prohibit regular network flow; disrupting services of the dataset is classified using a clustering method. Therefore, researchers used several ML and DL-based intrusion detection (ID) algorithms in different studies. NIDS uses a variety of machine learning algorithms and approaches to detect and differentiate between normal traffic, network attacks, and anomalies.

For the training and assessment of K-nearest neighbor (K-NN), Decision trees, Naive Bayes networks, artificial neural network (ANN) [6], and SVM [7], the majority of researchers use various IDS datasets. In [8], authors proposed an adaptive and resilient network intrusion detection system (IDS) for intrusion detection and classification of network attacks by using deep neural networks (DNNs). They utilized the UNSW-NB15 dataset to demonstrate the effectiveness model and achieved an overall significant accuracy. In [9], researchers also introduced an intrusion detection system based on artificial neural networks (ANN) and fuzzy clustering for zero-touch networks. This approach consists of three primary modules: fuzzy clustering, artificial neural networks (ANN), and fuzzy aggregation. A given set of data is divided into various clusters to perform classification. Using the ANN module, each subset's pattern is learned. The fuzzy aggregation module combines the findings of multiple ANNs to reduce detection errors in smart cities. Using the KDDCUP 99 dataset, the FC-ANN approach was examined and proved to be effective against low-frequency attacks such as remote to the user (R2L) and user to root (U2R). In addition, CICIDS2017 is realistic and utilizes up-to-date attacks [10]. In the [11] study, authors integrated smart contracts with eXplainable AI (XAI) to propose a robust cybersecurity framework for ZTN. The presented framework uses a blockchain and smart contract-enabled access control and authentication mechanism to build trust between the participating entities. Moreover, with the collected data, they introduced Digital Twins (DTs) to reproduce the attack detection operation in the ZTN environment and provide a forum for analysis. They used the Long Short-Term Memory (SALSTM) network to perform the attack detection capabilities of the proposed framework with the N-BaIoT and a self-generated DTs dataset. The novel IDS model has been proposed in [12] that was based on the Random Forest (RF), and Radial Basis Function Neural Network (RBFNN). They used an RF classifier for feature selection and the RBFNN algorithm for intrusion detection in CC environments. Additionally, Bot-IoT and NSL-KDD datasets are utilized to validate their proposed technique. This model obtained high accuracy (ACC) higher by utilizing the minimum features. Moreover, this model addressed the security challenges specific to CC environments and provided optimal solutions to increase the security in cloud systems. The study proposed a denial-of-service (DoS) attack-specific DL-based infiltration model. KDD CUP 1999 dataset (KDD), the most popular dataset for the assessment of intrusion, is utilized for the intrusion dataset [13]. This study focuses on the efficacy of machine and deep learning approaches in identifying cyber-attacks originating from compromised IoT devices and provides a thorough overview of IoT technology, risks, and intrusion detection models [14]. The author investigates the CICIDS-2017 dataset's effectiveness for evaluating IDS, highlighting its modern threat representation while addressing shortcomings through analysis and proposing an enhanced combined dataset for future intrusion detection engines [15]. The [16] study, presented a DL-based IDS model using Feed Forward Neural Networks

(FFNN), Random Neural Networks (RandNN), and Long Short-Term Memory (LSTM), to protect IoT networks from cyberattacks. This study performs experiments on the CIC-IoT22 dataset and achieves significant accuracy in detecting intrusion. The author investigates the problems in network security brought on by expanding data and new attacks [17]. The author introduces a novel stacked ensemble approach, utilizing base learners of ensemble models, in Web applications for anomaly-based intrusion detection [18]. The author described a unique stacked ensemble method for the detection of intrusions using anomalies in Web applications that makes use of base ensemble model learners [19].

In [20], network intrusion systems have evolved to inspect IP packets. NIDS now analyzes application layer headers and data for the filtration of IP datagram-based packets. To efficiently monitor application layer protocols such as HTTP and SMTP, more sophisticated security layers are needed to sift through the carried data. Different machine-learning techniques have been examined for the detection of streaming spam tweets. The study cited in [21] revealed that Naïve Bayes achieved an accuracy of 97.3%, surpassing other methods. The author described a deep learning-based strategy for creating such a productive and adaptable NIDS. Self-taught Learning (STL), a deep learning approach, is what we employ on the benchmark dataset for network intrusion, NSL-KDD. The study proposed a lightweight IDS for fog/IoT security that achieves great accuracy and recall on the ADFA-LD and ADFA-WD datasets [22]. The author discussed the security framework within the MonB5G project that employs SECaaS and AI for automated, distributed management of network slicing security in 5G and beyond, crucial for profitable services [23]. The [24] research, proposed a Non-Symmetric Deep Auto-Encoder System (NIDS) for network intrusion detection and validated its effectiveness and robustness using a benchmark KDD CUP'99 dataset. They implemented the model in the TensorFlow library and GPU framework and attained the highest accuracy. In the [16], authors designed a novel variational autoencoder (VAE) and attention-based gated recurrent units (AGRU)-based model for intrusion detection for ZTNS. Additionally, combining smart contracts (SCs), blockchain, and elliptic curve cryptography (ECG) through the authentication protocol to enhance secure data sharing in zero-touch networks.

While summarizing the literature review for successful network intrusion detection, the authors understand the value of examining not only IP packet headers but also application layer headers and contents. To better comprehend attack propagation and defense effectiveness, the studies concentrate on probabilistically modeling the spread of threats and applying critical reasoning. This is crucial for monitoring application layer protocols like HTTP and SMTP since they call for more advanced security measures to look at the data being transported. The detection of streaming spam tweets has been studied using several machine-learning techniques. In earlier research, the authors suggest deep learning-based solutions to answer the demand for a useful and flexible NIDS. The authors want to increase the NIDS's overall performance and adaptability while enhancing its capabilities.

The main focus of our study is to develop a network intrusion detection system for zero-touch networks inside smart cities that may detect various NIDS attacks and identify malicious traffic using deep learning models. We used the deep learning approach, such as Convolutional Neural Networks (CNNs) for instruction detection because it addresses multiple challenges that are posed by the essential requirement of manual feature extraction in other traditional approaches, like machine learning. The DL approaches allow the efficient processing of natural raw data and facilitate during the implementation, improving accuracy, and effectively identifying and stopping malicious or unauthorized activities. Recently, the Zero-Touch Network(ZTN) has emerged to automatically coordinate and manage network resources. ZTN is essential for operators to enhance by detecting harmful traffic that enables the detection of extensive traffic changes, hence influencing the physical system's characteristics. Enhancing the safety of smart cities and the compatibility, availability, and integrity of their services is the significance of this effort.

III. DATA AND PREPROCESSING

The proposed NIDS for zero-touch networks in smart cities is developed with a diverse structure, where several heterogeneous devices communicate with one another for the transmission of information. In smart cities, the devices detect and collect data from other devices using different policies and transmit the sensed data to the switches via complicated data flows, resulting in very dynamic and enormous IoT data traffic. With the extremely dynamic data traffic generated by communication in smart cities, the traffic load might also include malicious traffic. This results in a huge waste of channel resources and can also result in cyber-attacks. Hence there is a need for a model based on deep learning that can detect and filter out the malicious traffic in zero-touch networks and permits only normal traffic data exchange between the devices. In the next sections, a deep learning-based approach is discussed to solve the network traffic issue. The proposed deep learning-based NIDS model for zero-touch networks is shown in Figure 1.

The deep learning-based NIDS presents a promising approach to address the challenges of managing network traffic in smart cities. By connecting the power of deep convolutional neural networks, this solution contributes to building secure, efficient, and resilient zero-touch networks that support the growth and sustainability of smart cities while safeguarding against cyber threats. The trend of smart cities has led to a complex web of interconnected devices that communicate and exchange information, creating a dynamic and massive flow of IoT data traffic. This environment is inclined to both appropriate and malicious traffic, posing challenges for efficient network traffic management and cybersecurity. A deep learning-based approach is proposed to address these

TABLE 1. Comparative analysis of literature review.

Ref	Year	Data Set	Techniques	Results	Limitations
[9]	2020	KDD CUP 99	Deep learning-based techniques	-	Complex patterns are not considered
[14]	2019	UNSW-NB15	Recursive Feature Elimination and Random Forests	86.04% Accuracy	Different algorithms can be used for more accurate model evaluation
[15]	2019	KDD	J48, Random Forest, Decision Tree	93.7% Accuracy	False negative and false positive rates can be taken into consideration
[16]	2023	CICIDS-2018	Hybrid Deep-Learning-Based Network Intrusion Detection System (HDLNIDS)	98%	A limited number of features are utilized for model evaluation
[17]	2022	NID and BoT-IoT	CNN-based approach	92.85%	Different DL algorithms can be used for more accurate model evaluation
[18]	2020	CSIC-2010v2 and CICIDS-2017	Random forest, gradient boosting machine, and XGBoost.	99% Accuracy	The single dataset is used for model evaluation
[19]	2019	UNSW-NB15	Recursive Feature Elimination and Random Forests	86.04% Accuracy	Different algorithms can be used for more accurate model evaluation
[20]	2019	KDD	J48, Random Forest, Decision Tree	93.7% Accuracy	False negative and false positive rates can be taken into consideration
[21]	2019	ADFA-LD	Multilayer Perceptron (MLP) model for Fog computing	95.0% Accuracy	Fewer features are used for model evaluation
[22]	2021	MonB5G	Network slicing security orchestration framework	-	Real-world attack scenarios are missing.
[23]	2022	A real network that connected 12 cities in Poland.	ML-based techniques to improve ZSM	96% Accuracy	AI-generated attacks and targeted social engineering can ensure robust assessment.
[25]	2023	TON-IOT	Recurrent Neural Network and Gated Recurrent Units(RNN-GRU)	98%	The proposed system only relies on the ToN-IoT dataset, and not fully capture the range of attacks

TABLE 1. ***(Continued.)*** **Comparative analysis of literature review.**

[26]	2023	CSE-CIC-IDS 2018	Deep-learning-based techniques	93.74%	More relevant features can be extracted to enhance the output
[27]	2023	NSL-KDD and N-BaIoT	Machine learning-based technique	86.39%	Only focused on Rare-Class attack
[28]	2022	NSL-KDD	Attention-based CNN-BiLSTM	90.01%	Focused on a small number of attacks
[29]	2022	NSW-NB15 and NSL-KDD	Lightweight Boosting Algorithm	86% Accuracy	Integration of real-time threat intelligence feeds to enable proactive threat detection.
[30]	2023	KDD'99	Random forests algorithm	94.2% Detection Rate	Clustering algorithms are not considered which is important in such a problem.

challenges by effectively detecting and filtering out malicious traffic while permitting normal traffic to flow seamlessly within this heterogeneous, zero-touch network.

A. DATASET DESCRIPTION

The proposed research study used the CICIDS-2018 dataset that depicts real-time network behavior and includes a variety of intrusion modes. This dataset offers numerous intrusion patterns that can be used in the security sector with a range of network protocols and topologies. The CICIDS-2018 is a publicly available dataset that contains numerous patterns and intrusion methods. Table 2. shows the comparison of two datasets, i.e., CICIDS-2017 and CICIDS-2018. In CICIDS-2018 there are 80 statistical features from the sample that are measured in both forward and backward directions, such as packet length, volume, and the number of bytes. The CICIDS-2018 dataset is available in PCAP and CSV formats. The CICIDS-2018 dataset is a comprehensive repository of network traffic data that encompasses seven major categories of cyberattacks e.g., Brute-force attacks, which involve repeated attempts to gain unauthorized access to a system by guessing passwords or credentials. Additionally, the dataset covers Denial-of-Service (DOS) attacks, where attackers attack targeted systems with excessive traffic to disrupt their normal operation.

The Botnet category deals with data related to the control and activities of networks of compromised devices under the command of malicious actors. Heartbleed, another category, focuses on vulnerabilities like the Heartbleed bug in the OpenSSL cryptographic software library. Furthermore, Distributed Denial-of-Service (DDOS) attacks, which involve coordinated efforts to overwhelm a target's resources, are included in the dataset. Brute-force SSH attacks concentrate on unauthorized access attempts specifically aimed at Secure Shell (SSH) connections. Finally, the dataset also contains data related to Infiltration attempts, where attackers seek unauthorized access or information, and Web attacks, which involve a range of malicious activities targeting web applications and services. The diverse dataset provides valuable insights for researchers and cybersecurity professionals to study and mitigate various types of cyber threats. Comparisons are made between the sample sizes of the CICIDS-2018 dataset and the CICIDS-2017 dataset. The CICIDS-2018 dataset has a much larger sample size than the CICIDS-2017 dataset, notably in the Botnet and infiltration attacks, where CICIDS-2018 has increased by 143 and CICIDS-2017 by 4497, respectively.

B. DATA PREPROCESSING

The dataset consists of 13,500 instances that are either missing or have recursive feature values that cannot be processed, as determined by our investigation. The CICIDS-2018 dataset, which includes the system's runtime behavior and different intrusion mechanisms is available in PCAP and

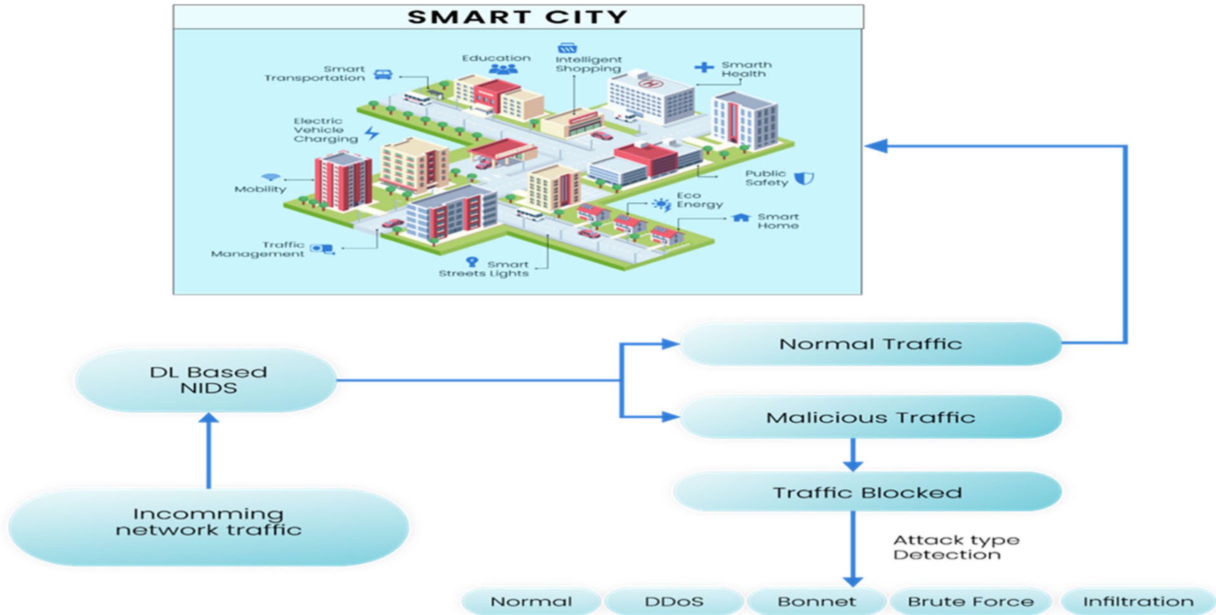

FIGURE 1. Deep learning-based NIDS for zero-touch network in smart city.

CSV formats. The dataset is about the dataset's structure, properties, and the seven attack categories. Identify and handle missing values in the dataset to ensure the quality of the data. To find and keep appropriate qualities for intrusion detection, analyze the relevance of attributes using methods like the Chi-square test. In the preprocessing phase, the features that provide limited or no contribution to the detection process are eliminated. Divide the preprocessed dataset into training and testing models to evaluate models as shown in Figure 2. To avoid any inconsistency, normalize numerical features to ensure they are on a similar scale, preventing some traits from being more dominant than others. The values in Table 2 describe how often various attack types are in the datasets. Compared to the CICIDS-2017 dataset, the CICIDS-2018 dataset, for example, has many more instances in most categories, indicating a more varied and comprehensive picture of network behavior and attack scenarios. It's also crucial to notice that the "Port Scan" category instance counts are missing from the CICIDS-2018 dataset. This can be because of changes in data-gathering methods or reporting practices between the two datasets. Different types of attacks have been categorized in the dataset. The categories include "Normal", "DDoS" (Distributed Denial of Service), "Brute Force", "Infiltration", "Dos" (Denial of Service), "Port Scan", "Web Attacks", and "Botnet". The records in each cell indicate the number of instances in the corresponding dataset that fall under a specific attack type.

The dataset's features in Table 3. are designed to capture different aspects of network flow behavior. These characteristics include inter-arrival periods between flows and packets, flow duration, packet quantity, packet size statistics, byte rates, and flags indicating certain packet qualities. The features reflect metrics such as average, standard deviation, maximum, and minimum values for time intervals and activity periods. These capabilities enable a thorough examination of network traffic and make it easier to spot and comprehend probable anomalies or intrusion patterns by providing insights into the dynamics of network communication, packet transmission patterns, and activity intervals.

The dataset comprises numerous recursive characteristics for classifying network activity. The proposed inquiry determines that the dataset is significantly unbalanced, and the use of machine learning techniques on an imbalanced dataset could result in majority class bias [15]. Therefore, the extremely imbalanced classes are removed, and a random forest is applied to reduce the size of the majority class. The selected features after preprocessing are shown in Table 3. The study has introduced the NIDS approach designed to tackle the mounting challenges posed by the exponential growth of network traffic and the need for robust cybersecurity mechanisms, particularly in the context of smart cities. A network of security cameras is normally used in smart cities to keep an eye on public areas, streets, and crucial infrastructure. CNNs are capable of being trained to recognize anomalies or strange activity, such as illegal access to permissible areas, or suspicious behavior [31]. It can also work well in the context of IoT where many linked devices are looking ahead for intrusions. The proposed study is focused on transforming a dataset, consisting of N data points, into a set where anomalies can be identified in network traffic. Training a CNN for intrusion detection requires a dataset whereas in the proposed study CICIDS-2018 dataset is used.

TABLE 2. CICIDS-2018 ID and CICIDS-2017 dataset evaluation.

Dataset	Normal	DDoS	Brute Force	Infiltration	Dos	Port Scan	Web Attacks	Botnet
CICIDS-2017	1,743,179	128,027	13,835	36	252,661	158,930	2180	1966
CICIDS-2018	6,112,151	687,742	380,949	161,934	654,301	-	928	286,191

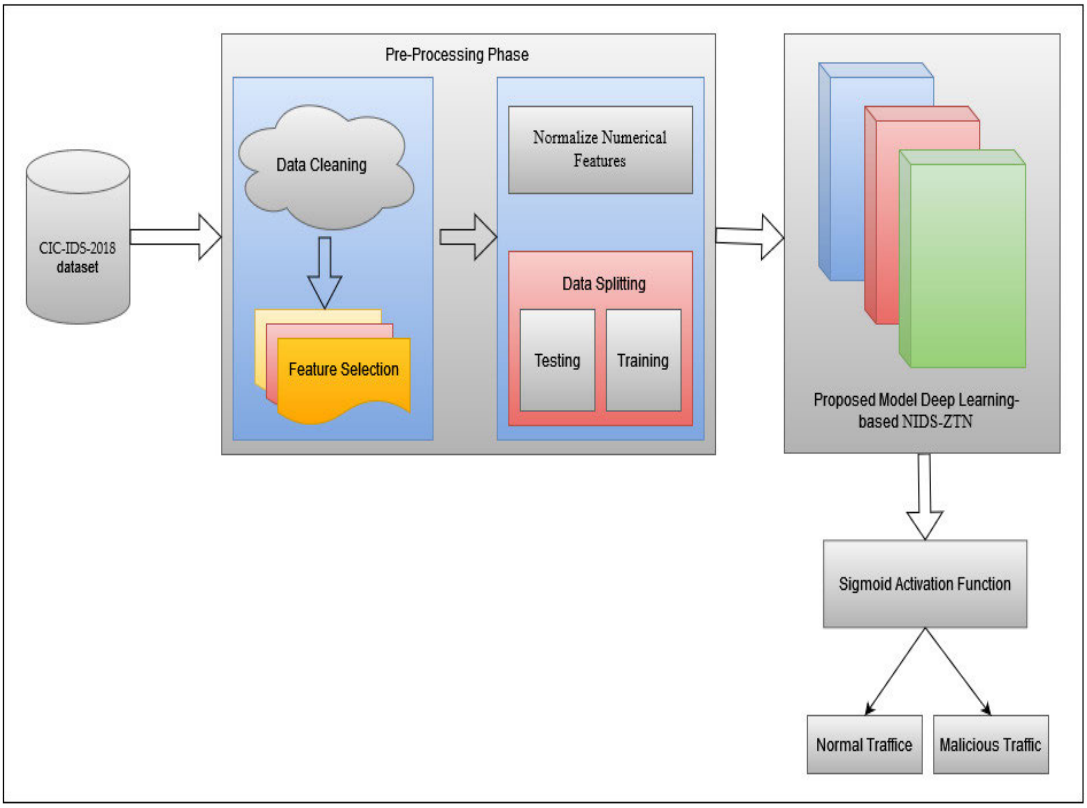

FIGURE 2. Proposed DL-based NIDS-ZTNS for intrusion detection in smart city.

IV. METHODOLOGY

A. PROPOSED DEEP LEARNING-BASED NIDS

The rapid increase in network traffic has come with the need for an effective and accurate intrusion detection system. The researcher has provided a comprehensive overview of the proposed deep learning-based network intrusion detection system. A novel approach has been proposed utilizing a deep convolutional network for an enhanced intrusion detection system. Deep learning has demonstrated tremendous promise in its capacity to extract hierarchical features from data. The proposed study has explored the potential of a convolutional neural network in the context of network traffic analysis for detection and investigates our proposed NIDS model. Then to assure the quality and sustainability of training and evaluation a wide data set is used which is processed using data cleaning feature extraction and normalization. The model has been trained using different methods e.g., dataset splitting, model initialization, and hyperparameters to maximize its performance. Our system used high-tech techniques for intrusion detection methods, incorporating cutting-edge methodologies and heuristics for reliable detection. The entire process is illustrated in Figure 3.

The proposed model is based on 1-D CNNs which refers to a network that has 5 layers stacked in a sequence and analyzes network traffic data to automatically learn discriminative features from raw traffic data, and capture patterns and anomalies indicative of attacks. Moreover, it takes in sequential data and processes them through convolutional layers to extract relevant features. After that, these features are passed through dropout layers, and max-pooling layers to reduce the dimensionality(downsampling), and flattening layer and capture the most important information. Finally, the network uses fully connected layers to classify the traffic as normal or malicious and produce the result on the output layer. The raw data, like network traffic data or log data, is fed into the network at the input layer. Convolutional layers initially create a series where each layer captures various features and patterns by applying a set of trainable filters to the input data.

TABLE 3. CICIDLS-2018 ID dataset's features summary.

Features	Description
fl_dur	The duration of a data flow or link is its length, usually expressed in seconds
tot_fw_pk	Total forward packets are all the data packets sent in the forward direction
tot_bw_pk	The total number of data packets sent in the reverse direction is referred to as total backward packets
fw_pkt_l_std	The forward direction's packet length standard deviation
fl_byt_s flow	Flow bytes per second is a metric that calculates the rate at which data bytes are transmitted within a flow
fl_iat_avg	Flow inter-arrival time average
fl_iat_std	The spread or variation in the amount of time between successive flows is measured by the flow inter-arrival time standard deviation.
fl_iat_max	The maximum flow inter-arrival time represents the largest interval between two successive flows
fl_iat_min	The shortest amount of time between two consecutive flows is indicated by the flow inter-arrival time minimum
fw_iat_tot	The time added up between two packets arriving in the forward direction consecutively
fw_iat_avg	The mean time passed between two successive packets arriving in the forward direction
fw_iat_std	The variance in the amount of time between successive packets traveling in the forward direction is measured by the forward inter-arrival time standard deviation
bw_iat_min	Backward inter-arrival time minimum indicates the shortest time gap between two consecutive packets in the backward direction
fw_psh_flag	Push function is set in the TCP header of packets transmitted in the forward direction
fw_urg_flag.	Data that is urgent traveling in the forward direction
bw_urg_flag	Indicates the urgency of data in packets transmitted in the backward direction
atv_std	Activity standard deviation measures the variation in activity levels within a network flow or connection
atv_max	Activity maximum represents the highest level of activity observed within a network flow or connection
atv_min	The minimum level of activity within a network flow or link
idl_avg	The average amount of time that a network flow or link spends inactively
idl_std	The variance in idle time intervals within a network connection
idl_max	The greatest amount of idle time for a network flow
idl_min	The least amount of time that has been found idle within a network flow

We applied the default activation function of CNN named the ReLU activation function to add non-linearity by converting any negative input into zero which means the output is not proportional to the input. Moreover, it does not enable the neurons to get activated and reduce the vanishing gradient issue. Max pooling layers following a set of convolutional layers. Max pooling aids in reducing the feature maps' spatial dimensions, isolating the most important data, and enhancing

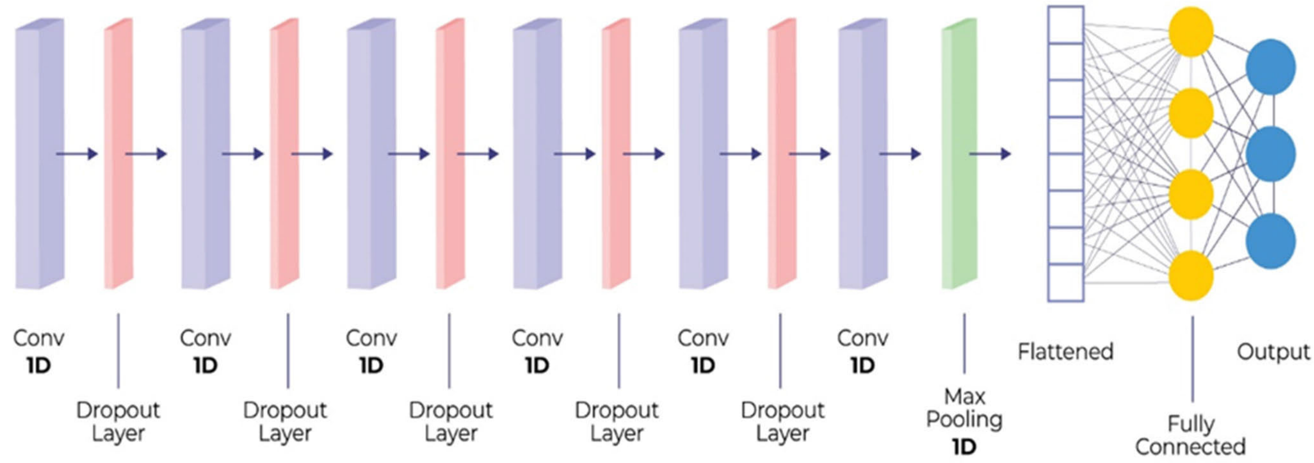

FIGURE 3. Proposed CNN-based deep learning architecture.

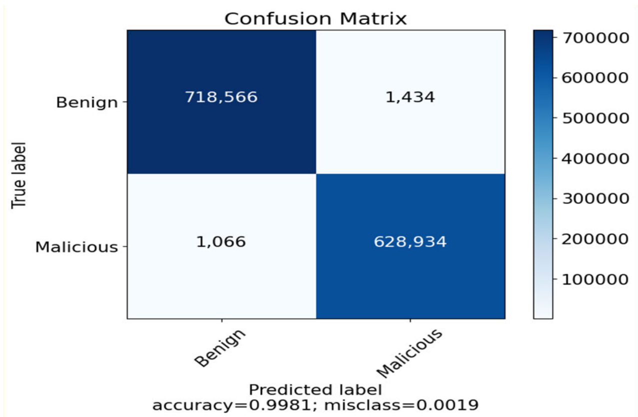

FIGURE 4. Binary intrusion detection confusion matrix.

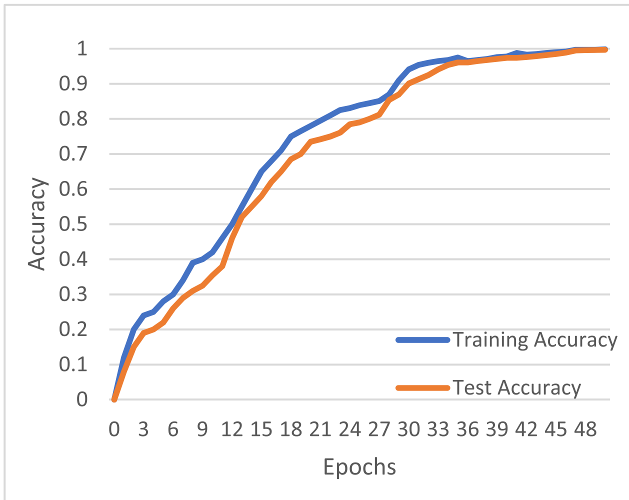

FIGURE 5. Accuracy graph.

TABLE 4. Model results.

Epoch	Accuracy	Loss	Precision	Recall	F1-Score
50	99.8	0.989	99.8	0.99	0.69

computing effectiveness. The dropout layers are added to avoid overfitting. During each forward and backward pass, dropout randomly sets a portion of the output units of the layer to zero. This motivates the network to pick up stronger and more widespread features. Convolutional layers, max-pooling layers, and dropout layers are repeated several times. This aids the network's learning of hierarchical features that are getting more complex.

The 2D feature maps are flattened into a 1D vector after the last set of convolutional and pooling layers. Whereas data is ready to be fed into the layers that are completely connected. Fully connected layers then connect the flattened features to a series of fully connected (dense) layers. After each fully connected layer, add dropout layers once more to reinforce regularization. Finally, we applied the sigmoid activation function to make up the final output layer. The sigmoid function is used to squeeze the output values between 0 and 1 that are gathered from CNN layers and detect either normal or intrusion activities since intrusion detection is a binary classification problem. The proposed technique is a deep-learning-based intrusion detection method, which uses past IoT traffic patterns to predict traffic loads. This adaptive channel assignment is conducted by deep learning using the training and testing data from the CICIDS-2018 dataset. The proposed model consists of five convolutional layers, where each layer is preceded by a dropout layer except the final convolutional layer which is followed by a max pooling layer. This layer is further connected to a fully connected layer that serves as an input for the output layer.

TABLE 5. Comparison of the proposed model with existing approaches.

Reference	Methodology	Datasets	Limitations	Accuracy
[32]	IDS based on ensemble learning for zero-touch networks	CSE-CIC-IDS2018	A diverse dataset should be considered for model evaluation	93%
[33]	CNN-LSTM	N-BaIoT	Only focuses on single Botnet attack type	90.88%
[34]	Anomaly-Based approaches	CICIDS2017	A smaller number of attacks is considered	-
[13]	Cu-LSTMGRU	CICIDS-2018	Limited features are used for model evaluation	98%
[35]	Hybrid CNN-LSTM	IoTID20	Only detect intrusions based on IoT infrastructure	98.80%
[36]	Deep learning techniques	NSL-KDD and UNSW-NB15	The single dataset is used for model evaluation	-
[37]	DL-based novel IDS for IoT networks	TON-IOT	The proposed model is not refined	98%
[38]	DL-ML-based SDN architecture	KDD-CUP99, and NSL-KDD	Only Explore SDN security and IDS about security concerns	98%
[39]	IoT-Based Smart City Framework	UNSW-NB15 and CICIDS2017	Only detect DoS and Bot attack	95%
[40]	Hybrid optimization and deep-learning-centric	NSL-KDD	IDS No dynamic methodology is considered	-
[41]	Deep learning-based model	Synthetic Replay Attack	Only detect replay attacks	97%
[42]	Deep learning-based approach	Distributed Smart Space Orchestration System traffic traces	Other algorithms can be tested	-
[43]	Hybrid ML algorithm	BoT-IoT	A hybrid model is not refined	-
[44]	Botnet detection system based on a two-level DL framework	BoT-IoT	Using anomaly detection and behavior analysis, Intrusion Detection Systems (IDS) are missing	96%

B. DISCUSSION

We have proposed an intrusion detection system based on a zero-touch network environment for monitoring the different types of malicious attacks. Furthermore, classify that network traffic into malicious network traffic or non-malicious network traffic. In the future, this could be useful in smart city environments with zero-touch network environments, where all networks might need to be set up without any human involvement. But deploying it there involves some ethical and privacy considerations that should be kept in mind, these are:

- Most importantly, the private information of sensitive network traffic should be secure by masking IP addresses, their location, and system passwords in the context of smart cities.
- Sharing data terms and conditions should be clear, including the purposes for which the data will be used, security measures to be implemented, and the responsibilities of each entity involved.
- Consider the ethical suggestions of data sharing, like potential risks to individuals' privacy and rights. Prioritize the interests and well-being of individuals when making decisions about data-sharing practices.
- Should use strong encryption techniques for data protection and ensure that only authorized persons can access the data, reducing the risk of unauthorized access or interception.

The network model can navigate the ethical and privacy challenges by implementing privacy-preserving mechanisms like differential privacy, data anonymization, federated learning, and privacy-aware feature selection. Differential privacy can generate noise to the data to protect the information of each user while maintaining the model's overall effectiveness. Federated learning ensures that sensitive data remains on local devices, sharing only model updates with the central system to minimize data breach risks. Data anonymization and aggregation further protect user privacy by deleting personally identifiable information. However, these strategies

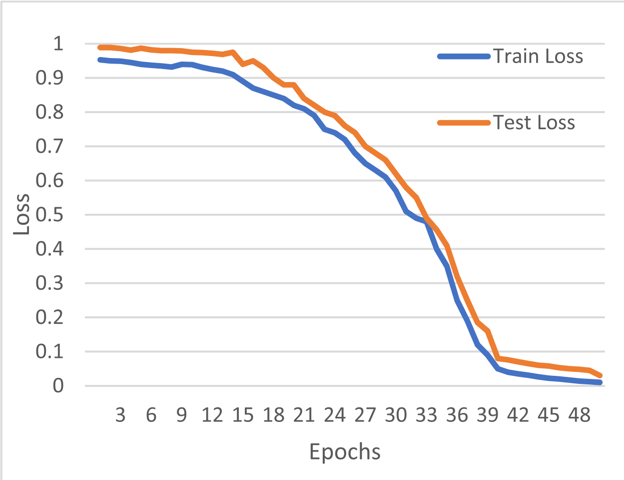

FIGURE 6. Loss graph.

comprise trade-offs, such as potential impacts on model accuracy or increased computational overhead. Despite these trade-offs, the system aims to balance security enhancement with privacy protection, ensuring robust intrusion detection while respecting user privacy within smart city environments.

C. SIMULATION SETUP

The proposed methodology used deep learning where Python is used to validate the presented NIDS approach. The experiment was carried out using clustered computers (Core i7, 64-bit, 24 GB RAM, 64-core CPU). The proposed model was developed using an NVIDIA GTX 1080 Ti GPU. To evaluate the performance of our proposed model, accuracy, loss training, and the testing dataset is used.

The CICIDS-2018 dataset is used to illustrate the efficacy of the model. The proposed approach distinguished between malicious and non-malicious data in network traffic. The proposed method reduces processing complexity to increase detection accuracy and reduce the loss graph.

D. PERFORMANCE METRICS AND RESULTS

Various performance measures such as accuracy and loss graphs are employed to assess the performance of the NIDS for zero-touch networks. The loss graph is defined as "how bad the proposed model is doing" concerning time. The effectiveness of NIDS for the zero-touch network is tested against the CICIDS_2018 data set.

The efficiency of the proposed model is measured using the metrics of accuracy and loss, precision, recall, and F1-score. The proposed model's performance is assessed through various metrics on a separate test set and by comparison with other established models on benchmark datasets. The training accuracy measures the proportion of correctly classified instances in the training dataset. In this case, the model achieved a high accuracy of 99.8%, indicating that it accurately predicted the majority of instances. Our model also obtained significant Precision, Recall, and F1-Score as 99.0%, 0.99%, and 0.695% respectively as mentioned in Table 4 calculated from the given confusion matrix in Figure 4. From the results as shown in Figure. 5 and Figure. 6, it is observed that the proposed model's accuracy keeps on increasing with time. This is because the proposed model can learn, which makes it more reliable and capable of differentiating between legitimate and malicious traffic. After the neural network has been trained on the training dataset, it gives an accuracy of 99.8%. We used a subset of the total dataset that contains Benign 718,566 and Malicious 628,934 samples respectively for the experiment. The used dataset has been divided into 30 and 70 ratios for the training and testing phase accordingly. These assessments demonstrate how well the model is doing on the data being used to train it as illustrated in Figure 4. The training loss value is 0.98 described as a measure of how well the model's predictions match the actual target values in the training dataset. A lower training loss is mentioned in Figure 5. indicates that the model's predictions are close to the actual values, which is a positive sign.

Our proposed novel methodology is called DL-NIDS-ZTN (Zero-Touch Networks) which promises the highest accuracy of 99.8% on the CICDS-2018 dataset as compared to the existing studies as mentioned in Table 5.

V. CONCLUSION

This study introduces a novel DL-based approach to address the challenges posed by the high volume of sensing data and network traffic within smart cities. As these cities evolve into interconnected and data-driven ecosystems, the need for effective intrusion detection systems becomes utmost to protect against potential cyber threats that can exploit the growing network infrastructure. For long-term success and to prevent intrusions, zero-touch networks in the smart industrial environment must be protected from cyberattacks. An effective and accurate IDS is required to ensure the zero-touch network operates at its peak functionality. The chance of malicious traffic entering the system increases as network traffic increases, making the smart city more susceptible to various cyberattacks. To create an IDS, this study suggests a multi-layered approach based on how the systems can be prevented from intrusions. Feature selection is performed to extract the most relevant features from the CICIDS-2018 dataset. The suggested DL-NIDS-ZTN method utilizes deep learning to act as intelligent protection, identifying and classifying network traffic to allow only valid data flows within the smart city infrastructure. This study introduces a novel methodology called DL-NIDS-ZTN (Zero-Touch Networks) which promises the highest accuracy of 99.8%. The thorough evaluation of the deep learning-based model on the CICIDS-2018 dataset validates its effectiveness and demonstrates how well it performs in strengthening the security of smart cities against new cyberattacks.

Future research can focus on refining real-time threat detection e.g., by utilizing the quick development of deep learning techniques, it is possible to explore ongoing improvements and changes to the proposed DL-NIDS-ZTN model. Additionally, the integration of real-time functionalities and tools capabilities could further elevate the system's effectiveness in rapidly identifying threats.

CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

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